

Transposes, Permutations, Vector Spaces

Finish elimination with $PA = LU$, flip matrices with transposes, then step up from vectors to whole spaces of vectors, and the column space of a matrix.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. Permutations and $PA = LU$

Row exchanges as matrices, all six 3×3 permutations, and why $P^{-1} = P^T$.

2. Transposes

Rows become columns. Portfolio return as $r^T x$, and $(AB)^T = B^T A^T$ entry by entry.

3. Symmetric matrices and $R^T R$

Why $R^T R$ is always symmetric, and how covariance matrices come from it.

4. Vector spaces

A set you can add and scale inside without leaving: portfolios as a space.

5. When a set is not a space

Long-only portfolios, fixed-budget portfolios and unions fail the test.

6. Subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$

Lines and planes through the origin. Zero-cost (self-financing) trades form one.

7. The column space $C(A)$

All combinations of the columns: exactly the targets your funds can reach.

METHODS IN THIS LESSON

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PART 2 Prerequisites

Click any link to review it, then come back.

- Span and column space, first look (Lesson 1): "every point you can reach by mixing".
- Permutation matrices (Lesson 2): PA swaps rows, AP swaps columns.

- $PA = LU$ and transposes of products (Lesson 4).
- The entry rule and building-block trades (Lesson 3).

PART 3 Explanation

THE RUNNING EXAMPLE: THE SAME FUNDS, NEW QUESTIONS

Funds: Nifty (12%, beta 1.0), bonds (7%, beta 0), midcap (15%, beta 1.2). Rules: budget, return (₹ thousand), market exposure. The fund matrix and the matrix of **scenario returns** we'll use:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \quad R = \begin{bmatrix} -10 & 4 \\ 10 & 7 \\ 30 & 10 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{bad year} \\ \leftarrow \text{normal year} \\ \leftarrow \text{good year} \end{array}$$

R lists the yearly return (%) of Nifty (column 1) and bonds (column 2) in three scenarios. This lesson asks three new questions. What happens when rows and columns swap roles? Which sets of portfolios behave like a "space"? And exactly which targets can a set of funds reach?

1. Permutations and $PA = LU$

Strang starts by finishing Chapter 2. Elimination sometimes meets a 0 in a pivot position and must exchange two rows (Lesson 2). Lesson 4 wrote that as $PA = LU$. Here we take it apart slowly: what a permutation matrix is, why we need one, how it fits into $PA = LU$, and how to solve with it.

What a permutation matrix is

A permutation matrix P is the identity matrix with its rows put in a different order. Multiplying PA reorders the rows of A in exactly that way. The easy way to read P : **each row of P is an instruction**. A 1 in column j means "take row j of A ".

1 BUILD IT

To swap rows 2 and 3, swap rows 2 and 3 of the identity: $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

2 READ IT

Row 1 of P_{23} is $(1, 0, 0)$: "take row 1". Row 2 is $(0, 0, 1)$: "take row 3". Row 3 is $(0, 1, 0)$: "take row 2".

WHY THAT WORKS

By the row way of multiplying (Lesson 3), row i of $PA = (\text{row } i \text{ of } P) \times A = \text{a combination of the rows of } A$. A row like $(0, 0, 1)$ takes 0 of row 1, 0 of row 2 and 1 of row 3, which is just row 3.

Why we need one: the credit fund

Lesson 2's **credit fund** earns 12% like Nifty but has beta 0. With Nifty, credit and midcap, the fund matrix and a target are

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 12 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \begin{array}{l} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{array} \quad b = \begin{bmatrix} 10 \\ 126 \\ 7.4 \end{bmatrix}.$$

1 WHAT WE DO

HOW

Start elimination in the usual order.

$\ell_{21} = 12$: row 2 $- 12 \times$ row 1 $= (12 - 12, 12 - 12, 15 - 12) = (0, 0, 3)$. $\ell_{31} = 1$: row 3 $-$ row 1 $= (1 - 1, 0 - 1, 1.2 - 1) = (0, -1, 0.2)$.

WHAT WE HAVE NOW

$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 3 \\ 0 & -1 & 0.2 \end{bmatrix}$: a 0 in the second pivot position.

2 WHY IT HAPPENED

THE FIX

Nifty and the credit fund both earn 12 per lakh. Removing Nifty from the return rule ("return $- 12 \times$ budget") removed the credit fund too. Below that 0 sits -1 , which is not zero, so exchange rows 2 and 3. If we had done the exchange **before** starting, elimination would never have stopped. That is the idea of $PA = LU$: reorder first, then eliminate without interruption.

RULE · $PA = LU$

For every invertible A there is a permutation P that puts the rows in a workable order, and then $PA = LU$ with no further exchanges: L holds the multipliers, U the pivots. When no exchange is needed, $P = I$ and this is the familiar $A = LU$.

PA = LU for the credit fund, entry by entry

1 STEP 1: REORDER

$P_{23}A$ takes row 1, then row 3, then row 2 of A :

RESULT

$$P_{23}A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1.2 \\ 12 & 12 & 15 \end{bmatrix}: \text{rows are now budget, exposure, return.}$$

2 STEP 2: COLUMN 1

$$\ell_{21} = 1/1 = 1: \text{row 2} - \text{row 1} = (1 - 1, 0 - 1, 1.2 - 1) = (0, -1, 0.2). \ell_{31} = 12/1 = 12: \text{row 3} - 12 \times \text{row 1} = (12 - 12, 12 - 12, 15 - 12) = (0, 0, 3).$$

3 STEP 3: COLUMN 2

The entry below the second pivot -1 is already 0, so $\ell_{32} = 0/(-1) = 0$: nothing to do.

WHAT WE HAVE NOW

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0.2 \\ 0 & 0 & 3 \end{bmatrix} \text{ (pivots } 1, -1, 3) \text{ and, with the multipliers in place,}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 12 & 0 & 1 \end{bmatrix}.$$

4 STEP 4: CHECK $LU = PA$

Row by row of L (each row of LU is a combination of the rows of U):

ROW 1

$$1 \times (1, 1, 1) = (1, 1, 1) \checkmark \text{ (budget).}$$

ROW 2

$$1 \times (1, 1, 1) + 1 \times (0, -1, 0.2) = (1, 1 - 1, 1 + 0.2) = (1, 0, 1.2) \checkmark \text{ (exposure).}$$

ROW 3

$$12 \times (1, 1, 1) + 0 \times (0, -1, 0.2) + 1 \times (0, 0, 3) = (12, 12, 12 + 3) = (12, 12, 15) \checkmark \text{ (return).}$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{P_{23}} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 12 & 12 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}}_A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 12 & 0 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0.2 \\ 0 & 0 & 3 \end{bmatrix}}_U$$

Solving with $PA = LU$: reorder b too

$Ax = b$ and $PAx = Pb$ are the same equations listed in a different order. So **apply the same reordering to b** , then do the two triangular solves of Lesson 4.

1 REORDER B

Pb takes entry 1, then entry 3, then entry 2: $(10, 126, 7.4) \rightarrow (10, 7.4, 126)$.

2 FORWARD: $LC = PB$

$$c_1 = 10. \quad 1 \cdot c_1 + c_2 = 7.4, \text{ so } c_2 = 7.4 - 10 = -2.6. \quad 12 \cdot c_1 + 0 \cdot c_2 + c_3 = 126, \text{ so } c_3 = 126 - 120 = 6.$$

3 BACK: $UX = C$

Row 3: $3z = 6$, so $z = 2$. Row 2: $-y + 0.2(2) = -2.6$, so $-y = -3$ and $y = 3$. Row 1: $x + 3 + 2 = 10$, so $x = 5$.

ANSWER

₹5 lakh Nifty, ₹3 lakh credit fund, ₹2 lakh midcap, the same as Lesson 2. Check in the original rules: budget $5 + 3 + 2 = 10$ ✓; return $60 + 36 + 30 = 126$ ✓; exposure $5 + 0 + 2.4 = 7.4$ ✓.

Since $P^{-1} = P^T$ (below), the factorization can also be written with A alone: $A = P^T LU$.

Strang's practical note: small pivots

Algebra only **requires** an exchange when a pivot is exactly 0. Software such as MATLAB also exchanges when a pivot is merely **small**, because a tiny pivot means a huge multiplier, and a huge multiplier blows up rounding errors. A two-line example: $0.0001x + y = 1$ and $x + y = 2$, whose true solution is $x \approx 1.0001$, $y \approx 0.9999$.

	No exchange (pivot 0.0001)	Exchange first (pivot 1)
Multiplier	$1/0.0001 = 10,000$	$0.0001/1 = 0.0001$
y , rounded to 3 decimals	1.000	1.000
Back-substitute for x	$x = (1 - y)/0.0001 = 0/0.0001 = 0$ ✗	$x = 2 - y = 1.000$ ✓

The same rounding of y is harmless in one order and destroys x in the other. Choosing the largest available pivot in each column is called **partial pivoting**; it is why software may swap rows even when the algebra doesn't need it.

How many permutation matrices are there?

Count the ways to reorder 3 rows. The new top row can be any of the 3 rows; the next can be either of the 2 left; the last is whatever remains: $3 \times 2 \times 1 = 6$. In general there are $n! = n(n - 1) \cdots 2 \cdot 1$ permutation matrices of size n (leaving the order alone counts as one).

Matrix	Rows of PA are rows... of A	What it does	Its inverse
$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$	1, 2, 3	nothing	itself
$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = P_{12}$	2, 1, 3	swap rows 1 and 2	itself
$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = P_{13}$	3, 2, 1	swap rows 1 and 3	itself
$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = P_{23}$	1, 3, 2	swap rows 2 and 3	itself
$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$	2, 3, 1	cycle: every row moves up one, the top goes to the bottom	the cycle below
$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$	3, 1, 2	cycle the other way	the cycle above

For larger sizes the count explodes: $4! = 24$ and $5! = 120$ ("listing all the five by five permutations would be not so much fun", Strang).

A closed family. Reordering twice is still a reordering, so the product of two permutations is again on the list. Example, $P_{12}P_{23}$: first P_{23} swaps rows 2 and 3, then P_{12} swaps rows 1 and 2. Multiplying out gives $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, the second cycle. In the other order, $P_{23}P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, the first cycle. Different again: even permutations don't commute. A family closed under multiplication and inverses is what mathematicians call a *group*.

Why $P^{-1} = P^T$, worked for a cycle

RULE · $P^{-1} = P^T$

The inverse of a permutation matrix is its transpose: $P^T P = I$. Undoing a reordering is the reverse reordering, and transposing P produces exactly that.

1 TAKE

the cycle $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$: new row 1 = old row 2, new row 2 = old row 3, new row 3 = old row 1.

TRANSPOSE

Rows become columns: $P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

2 MULTIPLY

$P^T P$, entry by entry. Row 1 of P^T is $(0, 0, 1)$; dotted with the columns of P , $(0, 0, 1)$, $(1, 0, 0)$, $(0, 1, 0)$: $1, 0, 0$. Row 2, $(1, 0, 0)$: $0, 1, 0$. Row 3, $(0, 1, 0)$: $0, 0, 1$.

WHAT WE HAVE NOW $P^T P = I$ ✓: each 1 in P^T lands exactly on a 1 in P ("the ones hit the ones", Strang).

IN ONE LINE

Row exchanges are matrices; $PA = LU$ covers every invertible A ; there are $n!$ permutations and each has $P^{-1} = P^T$.

▼ **Check yourself (MIT problem 5.1):** find a 3×3 permutation with $P^3 = I$ but $P \neq I$.

The cycle above. P moves each row one step round a cycle of length 3, so after three applications every row is back where it started: $P^3 = I$. (Multiplying out: $P^2 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = P^T$, and $P^2 P = P^T P = I$.)

2. Transposes

METHOD · TRANSPOSE

The transpose A^T flips A across its main diagonal: rows become columns and columns become rows. In symbols, $(A^T)_{ij} = A_{ji}$: the entry in row i , column j of A^T is the entry in row j , column i of A . An $m \times n$ matrix has an $n \times m$ transpose.

Transposing the fund matrices

1 2×2

$A = \begin{bmatrix} 1 & 12 \\ 1 & 7 \end{bmatrix}$. Row 1 $(1, 1)$ becomes column 1; row 2 $(12, 7)$ becomes column 2: $A^T = \begin{bmatrix} 1 & 12 \\ 1 & 7 \end{bmatrix}$.

CHECK ONE ENTRY $(A^T)_{12}$ should be $A_{21} = 12$ ✓.

2 3×3

RESULT

Each row of A becomes a column:

$A^T = \begin{bmatrix} 1 & 12 & 1 \\ 1 & 7 & 0 \\ 1 & 15 & 1.2 \end{bmatrix}$. Diagonal entries $1, 7, 1.2$ stay put; $A_{21} = 12$ moved to position $(1, 2)$, $A_{32} = 0$ to $(2, 3)$, and so on.

3 RECTANGULAR

With only Nifty and bonds but all three rules, A is 3×2 : $\begin{bmatrix} 1 & 1 \\ 12 & 7 \\ 1 & 0 \end{bmatrix}$. Its transpose is 2×3 : $\begin{bmatrix} 1 & 12 & 1 \\ 7 & 0 \end{bmatrix}$.

MEANING

In A each **column** is a fund; in A^T each **row** is a fund's full profile: Nifty = $(1, 12, 1)$, bonds = $(1, 7, 0)$.

Row vectors and the dot product

Vectors are columns. Their transposes are rows. That gives a compact way to write a dot product:

METHOD · DOT PRODUCT AS $x^T y$

For column vectors x and y with n entries, $x^T y$ is a $(1 \times n)(n \times 1) = 1 \times 1$ product: a single number, $x_1 y_1 + \dots + x_n y_n$.

Portfolio return in one line. Let $r = (12, 7, 15)$ be the returns per lakh and $x = (5, 3, 2)$ the holdings. Then

$$r^T x = \begin{bmatrix} 12 & 7 & 15 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} = 12 \times 5 + 7 \times 3 + 15 \times 2 = 60 + 21 + 30 = 111.$$

₹111 thousand: exactly the return rule (row 2 of A) applied to client 1. Finance writes portfolio return as $w^T r$ ("weights transpose returns") all the time; this is that formula.

The transpose of a product reverses the order

RULE · $(AB)^T = B^T A^T$

Transpose each factor and reverse the order, just like inverses (Lesson 4). Also $(A^T)^T = A$: flipping twice returns the original.

Check it on Lesson 3's two clients, $A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$ and $X = \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix}$, where $AX = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix}$:

1 LEFT SIDE

Transpose AX : $(AX)^T = \begin{bmatrix} 10 & 105 \\ 10 & 90 \end{bmatrix}$.

2 RIGHT SIDE

$X^T = \begin{bmatrix} 7 & 3 \\ 4 & 6 \end{bmatrix}$ and $A^T = \begin{bmatrix} 1 & 12 \\ 7 & 0 \end{bmatrix}$. Entry by entry: $(7 \cdot 1 + 3 \cdot 1, 7 \cdot 12 + 3 \cdot 7) = (10, 105)$ for row 1, and $(4 \cdot 1 + 6 \cdot 1, 4 \cdot 12 + 6 \cdot 7) = (10, 90)$ for row 2.

WHAT WE HAVE NOW $X^T A^T = \begin{bmatrix} 10 & 105 \\ 10 & 90 \end{bmatrix} = (AX)^T \checkmark$.

3 MEANING

AX had one **column** per client; $(AX)^T$ has one **row** per client: (budget, return) for client 1, then client 2. That's the same report sorted the other way, like a spreadsheet with clients as rows.

WHY NOT $A^T X^T$?

$A^T X^T = \begin{bmatrix} 1.7+12.4 & 1.3+12.6 \\ 1.7+7.4 & 1.3+7.6 \end{bmatrix} = \begin{bmatrix} 55 & 75 \\ 35 & 45 \end{bmatrix}$: wrong order, meaningless numbers.

IN ONE LINE

$(A^T)_{ij} = A_{ji}$; $x^T y$ is a dot product; $(AB)^T = B^T A^T$.

▼ Check yourself (Strang's example): transpose $\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$.

Column 1 (1, 2, 4) becomes row 1 and column 2 (3, 3, 1) becomes row 2: $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}$, a 2×3 matrix.

3. Symmetric matrices and $R^T R$

DEFINITION: SYMMETRIC MATRIX

A square matrix is **symmetric** if transposing doesn't change it: $S^T = S$, that is $S_{ij} = S_{ji}$ for every i, j . Everything above the diagonal is mirrored below it. Example: $\begin{bmatrix} 3 & 1 & 7 \\ 1 & 2 & 9 \\ 7 & 9 & 4 \end{bmatrix}$.

Strang: symmetric matrices "show up in a lot of applications", and he shows where they come from. Take **any** matrix R , even a rectangular one, and multiply its transpose by it.

RULE · $R^T R$ IS ALWAYS SYMMETRIC

For any $m \times n$ matrix R , the $n \times n$ matrix $R^T R$ is symmetric. Proof in two lines: $(R^T R)^T = R^T (R^T)^T = R^T R$, using "reverse the order" and $(R^T)^T = R$. (RR^T , which is $m \times m$, is symmetric for the same reason.)

$R^T R$ for the scenario returns, entry by entry

R is 3×2 (3 scenarios $\times$ 2 funds), so $R^T R$ is 2×2 . Entry (i, j) = (column i of R) $\cdot$ (column j of R). The columns are Nifty $(-10, 10, 30)$ and bonds $(4, 7, 10)$.

1 ENTRY (1, 1)

Nifty $\cdot$ Nifty = $(-10)^2 + 10^2 + 30^2 = 100 + 100 + 900 = 1100$.

2 ENTRY (1, 2)

Nifty · bonds = $(-10)(4) + (10)(7) + (30)(10) = -40 + 70 + 300 = 330$.

ENTRY (2, 1)

bonds · Nifty: the same three products in the other order, $4(-10) + 7(10) + 10(30) = 330$. **This is the symmetry:** a dot product doesn't care which vector comes first.

3 ENTRY (2, 2)

bonds · bonds = $4^2 + 7^2 + 10^2 = 16 + 49 + 100 = 165$.

WHAT WE HAVE NOW $R^T R = \begin{bmatrix} 1100 & 330 \\ 330 & 165 \end{bmatrix}$: symmetric ✓.

The 3×3 version RR^T compares **scenarios** instead of funds (entry (i, j) = row i · row j of R):

$$RR^T = \begin{bmatrix} (-10)^2 + 4^2 & (-10)(10) + 4 \cdot 7 & (-10)(30) + 4 \cdot 10 \\ \cdot & 10^2 + 7^2 & 10 \cdot 30 + 7 \cdot 10 \\ \cdot & \cdot & 30^2 + 10^2 \end{bmatrix} = \begin{bmatrix} 116 & -72 & -260 \\ -72 & 149 & 370 \\ -260 & 370 & 1000 \end{bmatrix}$$

Only the upper triangle needed computing; symmetry fills in the rest.

MODEL · COVARIANCE MATRIX (PREVIEW)

Subtract each fund's average return first, then form $\frac{1}{m} C^T C$ (with m scenarios): that is the **covariance matrix** of the funds, the core input of portfolio risk models. Because it has the form $C^T C$, a **covariance matrix is always symmetric**.

1 AVERAGES

Nifty: $(-10 + 10 + 30)/3 = 10$. Bonds: $(4 + 7 + 10)/3 = 7$.

SUBTRACT THEM

Nifty: $(-10 - 10, 10 - 10, 30 - 10) = (-20, 0, 20)$. Bonds: $(4 - 7, 7 - 7, 10 - 7) = (-3, 0, 3)$. These are the columns of C .

2 $C^T C$

$(1, 1)$: $400 + 0 + 400 = 800$. $(1, 2) = (2, 1)$: $(-20)(-3) + 0 + 20 \cdot 3 = 60 + 60 = 120$. $(2, 2)$: $9 + 0 + 9 = 18$.

DIVIDE BY 3

Covariance = $\begin{bmatrix} 800/3 & 40 \\ 40 & 6 \end{bmatrix} \approx \begin{bmatrix} 266.7 & 40 \\ 40 & 6 \end{bmatrix}$.

MEANING

Nifty's variance ≈ 266.7 (volatility $\approx \sqrt{266.7} \approx 16.3\%$), bonds' variance 6 ($\approx 2.4\%$), and a positive covariance 40: in these scenarios the two funds move together. Risk models come back in the probability course.

IN ONE LINE

$S^T = S$ means mirrored across the diagonal; $R^T R$ is always symmetric, which is why covariance matrices are.

▼ Check yourself (MIT problem 5.2): how many entries of a 4×4 matrix can be chosen freely if it is symmetric? If it is skew-symmetric ($A^T = -A$)?

Symmetric: the 4 diagonal entries plus the 6 above the diagonal (the 6 below are copies): **10**.

Skew-symmetric: the diagonal must be 0 (since $a_{ii} = -a_{ii}$) and the lower triangle is minus the upper: **6**.

4. Vector spaces

Strang calls this "the beginning of real linear algebra": instead of single vectors, we look at whole spaces of vectors. What do we do with vectors? Two things: **add them** and **multiply them by numbers**. A space is a collection where those two operations never take you outside it.

RULE · VECTOR SPACE TEST (CLOSURE)

A collection V of vectors is a **vector space** if, for any v, w in V and any real number c : (1) $v + w$ is in V (closed under addition), and (2) $c v$ is in V (closed under multiplication by scalars). Together: every **linear combination** $c v + d w$ stays in V . (There are eight more rules about how addition behaves, such as $v + w = w + v$; for vectors of numbers they hold automatically.)

The big examples: $\mathbb{R}^n$

$\mathbb{R}^2$ is all vectors with 2 real components: the whole xy -plane. In portfolio terms, think of (x, y) as "₹ lakh in Nifty, ₹ lakh in bonds", where negative numbers mean selling short. Check the two rules on real portfolios:

1 ADD

Client 1 $(7, 3)$ plus client 2 $(4, 6)$, component by component: $(7 + 4, 3 + 6) = (11, 9)$. Merging two books gives another pair of numbers: still in $\mathbb{R}^2$ ✓.

2 SCALE

Double client 1: $2(7, 3) = (14, 6)$. Reverse it: $-1 \times (7, 3) = (-7, -3)$ (short both). Scale by zero: $0 \times (7, 3) = (0, 0)$, the empty

portfolio. All still in $\mathbb{R}^2$ ✓.

$\mathbb{R}^3$ is all vectors with 3 components, for example three-fund portfolios $(5, 3, 2)$, and $\mathbb{R}^n$ is all vectors with n components. Careful: $(3, 2, 0)$ is in $\mathbb{R}^3$, not $\mathbb{R}^2$. It has three components; one of them happens to be zero.

WATCH OUT: THE ZERO VECTOR IS ESSENTIAL

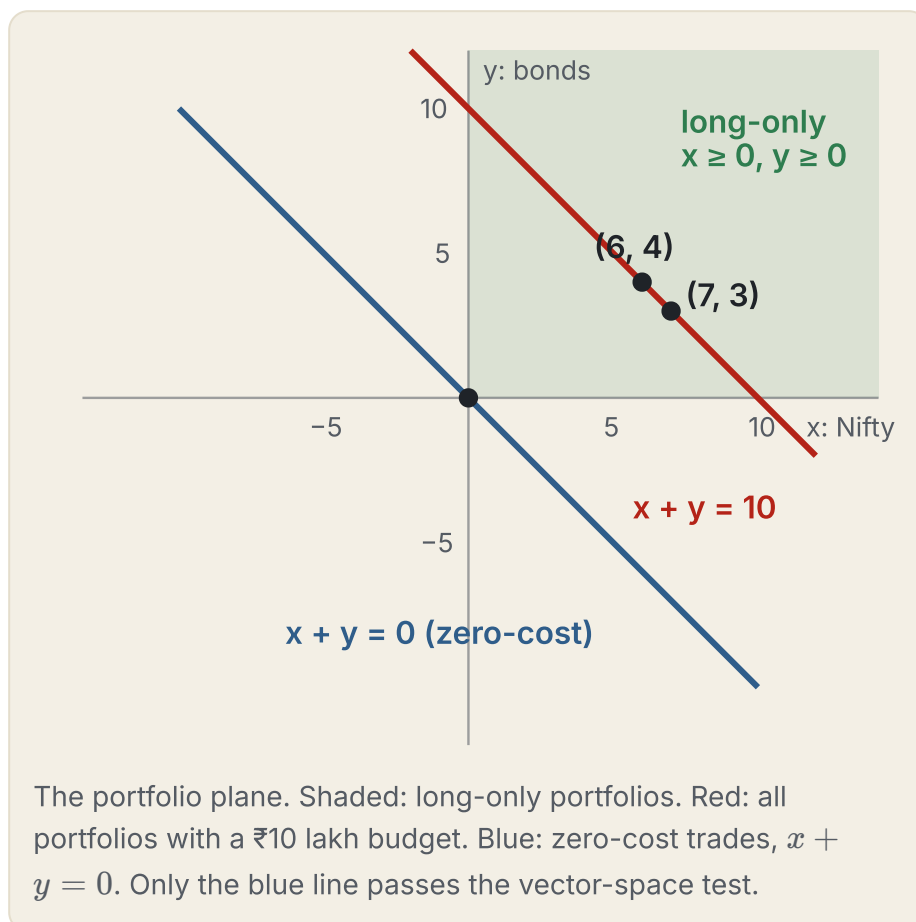
Every vector space contains 0 , because you must be allowed to multiply any vector by the number 0 . Strang: "Every vector space has got that zero vector in it." A plane with the origin removed is **not** a space.

IN ONE LINE

A vector space is closed under adding and scaling, so under all linear combinations, and always contains 0 .

5. When a set is not a space

The test is easiest to understand from sets that fail it. Here are three, all natural in portfolio work.



Failure 1: long-only portfolios (a quadrant)

Many funds may only **buy**, never sell short: $x \geq 0$ and $y \geq 0$. Strang uses exactly this quarter-plane as his first non-example.

1 ADDITION

Two long-only portfolios, $(7, 3) + (6, 4) = (13, 7)$: both components still ≥ 0 ✓. Adding is fine.

2 SCALING

Multiply $(7, 3)$ by -1 : $(-7, -3)$, both negative, so outside the quadrant ✗. (Strang multiplies by -5 ; any negative number fails.)

VERDICT

Closed under addition, **not** under scalar multiplication: **not a vector space**.

IN MARKETS

This is why a **long-only** constraint changes the maths of portfolio construction. Without it, the set of allowed portfolios is a vector space and linear algebra solves everything directly. With it, you need inequality-constrained optimisation, a different toolkit.

Failure 2: a fixed budget (a line not through the origin)

All portfolios that use exactly ₹10 lakh: $x + y = 10$ (the red line).

1 ADDITION

$(7, 3)$ and $(6, 4)$ both have budget 10. Their sum $(13, 7)$ has budget $13 + 7 = 20 \neq 10$ ✗.

2 ZERO

$0 \times (7, 3) = (0, 0)$ has budget $0 \neq 10$ ✗. The zero vector is not on the line.

VERDICT

Not a vector space. Strang: a line in $\mathbb{R}^2$ that misses the origin is never a subspace.

Failure 3: "all-Nifty or all-bonds" (a union of two lines)

From the recitation: take every all-Nifty portfolio $(t, 0)$ together with every all-bond portfolio $(0, t)$.

1 SCALING

A multiple of an all-Nifty portfolio is still all-Nifty, and the same for bonds ✓.

2 ADDITION

VERDICT

$(1, 0) + (0, 1) = (1, 1)$: a mixed portfolio, on neither line $\mathcal{X}$.

Not a **vector space**. Each line alone is a subspace, but their union isn't: adding escapes it.

IN ONE LINE

To fail, a set only has to let one sum or one multiple escape: long-only (negatives), fixed budget (sums and 0), unions (sums).

6. Subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$

RULE · SUBSPACE TEST

A **subspace** is a vector space sitting inside a bigger one. To check that a set inside $\mathbb{R}^n$ is a subspace: (1) it contains 0; (2) sums stay inside; (3) multiples stay inside. Strang: once a subspace contains a non-zero vector v , it must contain the **whole line** of multiples cv .

All the subspaces of $\mathbb{R}^2$

Subspace	Example in portfolio terms
All of $\mathbb{R}^2$	every portfolio of Nifty and bonds, long or short
A line through $(0, 0)$	all-Nifty portfolios $(t, 0)$; or zero-cost trades $(t, -t)$
Just $\{(0, 0)\}$	the empty portfolio alone

There is nothing in between. A line through the origin inside $\mathbb{R}^2$ looks like $\mathbb{R}^1$ but isn't the same thing: its vectors still have **two** components (Strang's point).

Zero-cost trades: a subspace that matters

MODEL · SELF-FINANCING (ZERO-COST) TRADES

A trade is **self-financing** (zero-cost) if the money it puts into some funds is exactly taken out of others: the changes add up to zero. Rebalancing an existing portfolio without adding or withdrawing money is always a zero-cost trade.

With two funds the zero-cost trades are $\{(x, y) : x + y = 0\}$, the blue line through $(1, -1)$. Run the subspace test:

1 ZERO

$(0, 0): 0 + 0 = 0 \checkmark$ ("do nothing" costs nothing).

2 SUMS

Two zero-cost trades $(a, -a)$ and $(b, -b)$ add to $(a + b, -a - b) = (a + b, -(a + b))$: components still add to 0 $\checkmark$. Example: $(2, -2) + (1, -1) = (3, -3)$.

3 MULTIPLES

$c(a, -a) = (ca, -ca)$: still adds to 0 $\checkmark$. Example: $-3 \times (2, -2) = (-6, 6)$.

VERDICT

A **subspace**: a line through the origin. Compare with "budget = 10": the only difference is the right-hand side, 0 instead of 10, and that 0 is what lets the origin in.

All the subspaces of $\mathbb{R}^3$

Subspace	Example with three funds
All of $\mathbb{R}^3$	every three-fund portfolio
A plane through 0	zero-cost trades: $x + y + z = 0$
A line through 0	zero-cost trades that also keep market exposure unchanged
Just $\{0\}$	the empty trade

Working out that line. A trade (x, y, z) that changes neither the budget nor the exposure needs $x + y + z = 0$ and $x + 1.2z = 0$.

1 FROM THE EXPOSURE

$$x = -1.2z.$$

RULE FROM THE BUDGET RULE

$$y = -x - z = 1.2z - z = 0.2z.$$

2 SO

every such trade is $z \times (-1.2, 0.2, 1)$. Taking $z = 5$ gives whole numbers: $(-6, 1, 5)$.

CHECK

Budget: $-6 + 1 + 5 = 0 \checkmark$. Exposure: $-6 + 1.2 \times 5 = -6 + 6 = 0 \checkmark$. Return: $12(-6) + 7(1) + 15(5) = -72 + 7 + 75 = 10$.

MEANING

The trades that change **only the return** form a line, and this one adds ₹10 thousand. It is exactly $10 \times$ the building-block column $(-0.6, 0.1, 0.5)$ of A^{-1} in Lesson 3.

That line is where the zero-cost plane and the zero-exposure plane meet. In general, **the intersection of two subspaces is a subspace** (both conditions survive), while a union usually isn't (topic 5). Add a third condition, "zero return change", and only $(0, 0, 0)$ is left: the three planes meet only at the origin, because the fund matrix is invertible.

IN ONE LINE

Subspaces are $\{0\}$, lines and planes through the origin, and the whole space. Zero-cost trades are a subspace; fixed-budget portfolios are not.

▼ **Check yourself (MIT problem 5.3): do the symmetric 2×2 matrices form a subspace of all 2×2 matrices? Do the non-symmetric ones?**

Symmetric: yes. If $A^T = A$ and $B^T = B$ then $(A + B)^T = A^T + B^T = A + B$ and $(cA)^T = cA^T = cA$; and the zero matrix is symmetric. Non-symmetric: no. $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, which is symmetric, so the sum escapes (and 0 isn't in the set at all).

7. The column space $C(A)$

Now the subspace that comes straight out of a matrix. Strang: take the columns, and "not satisfied with a few vectors, we want a space of vectors".

METHOD · COLUMN SPACE $C(A)$

The **column space** $C(A)$ of an $m \times n$ matrix is the set of **all linear combinations of its columns**. It is a subspace of $\mathbb{R}^m$: combinations of combinations are still combinations, and $0 \times$ every column gives 0.

Two funds, three rules: a plane in $\mathbb{R}^3$

Use only Nifty and bonds, but keep all three rules. Then A is 3×2 and its columns live in $\mathbb{R}^3$:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \\ 1 & 0 \end{bmatrix} \begin{array}{l} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{array} \quad C(A) = \left\{ x \begin{bmatrix} 1 \\ 12 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix} \right\}$$

Each choice of (x, y) gives one reachable target. All of them together fill a **plane through the origin** in $\mathbb{R}^3$: more than a line (the columns point in different directions), but not all of $\mathbb{R}^3$ (only two directions).

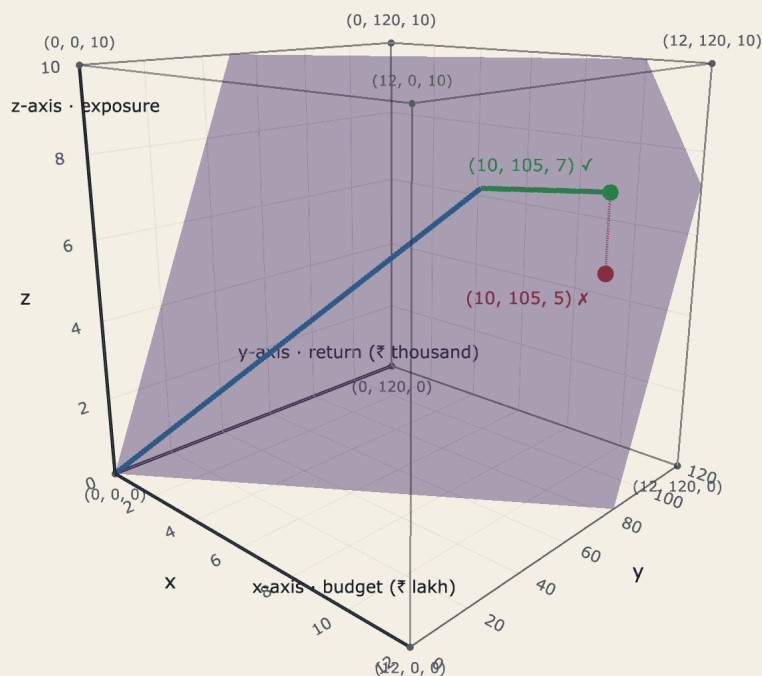
Which targets are in it?

Look at the exposure row: $1 \cdot x + 0 \cdot y = E$. With only Nifty (beta 1) and bonds (beta 0), the exposure always equals the Nifty amount. That one fact decides everything:

1	TARGET	$(10, 105, 7)$: ₹10 lakh, ₹105 thousand, exposure 7.
	HOW	Exposure row: $x = 7$. Budget row: $7 + y = 10$, so $y = 3$. Return row (the check): $12(7) + 7(3) = 84 + 21 = 105 \checkmark$.
	VERDICT	In $C(A)$: $7 \times (1, 12, 1) + 3 \times (1, 7, 0) = (10, 105, 7)$.

2	TARGET	$(10, 105, 5)$: the same budget and return, exposure 5.
	HOW	Exposure row: $x = 5$. Budget row: $y = 5$. Return row: $12(5) + 7(5) = 60 + 35 = 95 \neq 105 \times$.
	VERDICT	Not in $C(A)$: no mix of Nifty and bonds gives ₹105 thousand with exposure only 5. (You would need a third fund, such as midcap.)

The plane's equation. Every reachable (B, R, E) satisfies one equation: $-7B + R - 5E = 0$. Check it on both columns: Nifty $-7 + 12 - 5 = 0 \checkmark$, bonds $-7 + 7 - 0 = 0 \checkmark$. Then the targets: $(10, 105, 7)$ gives $-70 + 105 - 35 = 0$ (in), while $(10, 105, 5)$ gives $-70 + 105 - 25 = 10$ (out by 10). (The numbers $(-7, 1, -5)$ come from the *cross product* of the two columns, a 3D tool from the calculus course.)



■ $C(A)$: the plane $-7B + R - 5E = 0$ ■ 7 × Nifty (1, 12, 1) ■ + 3 × bonds (1, 7, 0)
 ● (10, 105, 7) ✓ in $C(A)$ ● (10, 105, 5) ✗ not in $C(A)$ ■■■ gap: exposure must be 7

The column space $C(A)$: a plane through the origin (drag to rotate). 7 Nifty steps plus 3 bond steps reach (10, 105, 7) on the plane. The target (10, 105, 5) sits off it: no mix of these two funds reaches it.

Three funds: the whole space

Add the midcap column (1, 15, 1.2). The three columns point in three independent directions (the matrix is invertible, Lesson 3), so $C(A)$ is all of $\mathbb{R}^3$: every (budget, return, exposure) target is reachable, which is why every client so far had an answer. With the balanced fund instead, the three columns lie in one plane and $C(A)$ is only that plane (Lesson 1).

WHAT'S NEXT

$Ax = b$ has a solution exactly when b is in $C(A)$. The next lesson turns this into the full story of when, and how many, solutions exist.

IN ONE LINE

$C(A)$ = all combinations of the columns = every target the funds can reach; it is a line, a plane or the whole space.

▼ Check yourself (Strang's example): what is the column space of $\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$? Is $(4, 5, 5)$ in it?

The plane through the origin containing $(1, 2, 4)$ and $(3, 3, 1)$ (they point in different directions, so it is a plane, not a line). And $(4, 5, 5) = 1 \times (1, 2, 4) + 1 \times (3, 3, 1)$, so yes: it is the sum of the two columns.